

● MEDIUM

● PROBLEM-SOLVING AND DATA ANALYSIS

● ANSWER KEY

Ratios, rates, proportional relationships, and units

01

10 answers · Rationale-rich · Teach from doc

Q1**B****B. 7,690**

Choice B is correct. The land area of the coastal city can be found by subtracting the area of the water from the total area of the coastal city; that is, $92.1 - 11.3 = 80.8$ square miles. The population density is the population divided by the land area, or $\frac{621,000}{80.8} = 7,686$, which is closest to 7,690 people per square mile.

Choice A is incorrect and may be the result of dividing the population by the total area, instead of the land area. Choice C is incorrect and may be the result of dividing the population by the area of water. Choice D is incorrect and may be the result of making a computational error with the decimal place.

Q2**A**

A. $\frac{4}{3}w$

Choice A is correct. It's given that rectangle A has length 15 and width w . Therefore, the length-to-width ratio of rectangle A is 15 to w . It's also given that rectangle B has length 20 and the same length-to-width ratio as rectangle A. Let x represent the width of rectangle B. The

proportion $\frac{15}{w} = \frac{20}{x}$ can be used to solve for x in terms of w . Multiplying both sides of this equation by x yields $\frac{15x}{w} = 20$, and then multiplying both sides of this equation by w yields

$15x = 20w$. Dividing both sides of this equation by 15 yields $x = \frac{20w}{15}$. Simplifying this

fraction yields $x = \frac{4}{3}w$.

Choices B and D are incorrect and may result from interpreting the difference in the lengths of rectangle A and rectangle B as equivalent to the difference in the widths of rectangle A and rectangle B. Choice C is incorrect and may result from using a length-to-width ratio of w to 15, instead of 15 to w .

Q3**A**

A. 7

Choice A is correct. It's given that the object has a mass of 168 grams and a volume of 24 cubic centimeters. Dividing the mass, in grams, of the object by the volume, in cubic centimeters, of the object gives the density, in grams per cubic centimeter, of the object. It follows that the density of the object is $\frac{168 \text{ grams}}{24 \text{ cubic centimeters}}$, which is equivalent to $\frac{168}{24}$ grams per cubic centimeter, or 7 grams per cubic centimeter.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q4**D**D. $146t$

Choice D is correct. It's given that the total mass, in kilograms, of r identical objects is t . To obtain the expression $146r$, r is multiplied by 146. Therefore, to find the total mass, in kilograms, of $146r$ of these objects, t must also be multiplied by 146. The result of multiplying t by 146 is the expression $146t$. Therefore, the total mass, in kilograms, of $146r$ of these objects, is $146t$.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Q5**2432**

The correct answer is 2,432. It's given that 4 cups = 1 quart. It follows that 76 quarts is equivalent to $76 \text{ quarts} \frac{4 \text{ cups}}{1 \text{ quart}}$, or 304 cups. It's also given that 8 fluid ounces = 1 cup. It follows that 304 cups is equivalent to $304 \text{ cups} \frac{8 \text{ fluid ounces}}{1 \text{ cup}}$, or 2,432 fluid ounces.

Q6**24**

The correct answer is 24. The equation $\frac{24x}{ny} = 4$ can be rewritten as $\frac{24x}{n} \cdot \frac{1}{y} = 4$. It's given that $\frac{x}{y} = 4$. Substituting 4 for $\frac{x}{y}$ in the equation $\frac{24x}{n} \cdot \frac{1}{y} = 4$ yields $\frac{24}{n} \cdot 4 = 4$. Multiplying both sides of this equation by n yields $244 = 4n$. Dividing both sides of this equation by 4 yields $24 = n$. Therefore, the value of n is 24.

Q7**A****A. $3(13.50z)$**

Choice A is correct. It's given that when Tanya works z hours, she earns $13.50z$ dollars. Since her hourly rate is constant, if she works 3 times as many hours, or $3z$ hours, she will earn 3 times as many dollars, or $3(13.50z)$.

Choice B is incorrect. This expression represents adding 3 dollars to the $13.50z$ dollars Tanya will earn. Choice C is incorrect. This expression can be rewritten as $16.50z$, which implies that Tanya earns \$16.50 per hour, not \$13.50. Choice D is incorrect. This expression adds 3 to the number of hours Tanya works, rather than multiplying the hours she works by 3.

Q8**B****B. It must increase by 24.5 units.**

Choice B is correct. It's given that the ratio of the rectangular region's length to its width is 35 to 10. This can be written as a proportion: $\frac{\text{length}}{\text{width}} = \frac{35}{10}$, or $\frac{l}{w} = \frac{35}{10}$. This proportion can be rewritten as $10l = 35w$, or $l = 3.5w$. If the width of the rectangular region increases by 7, then the length will increase by some number x in order to maintain this ratio. The value of x can be found by replacing l with $l + x$ and w with $w + 7$ in the equation, which gives $l + x = 3.5w + 7$. This equation can be rewritten using the distributive property as $l + x = 3.5w + 24.5$. Since $l = 3.5w$, the right-hand side of this equation can be rewritten by substituting l for $3.5w$, which gives $l + x = l + 24.5$, or $x = 24.5$. Therefore, if the width of the rectangular region increases by 7 units, the length must increase by 24.5 units in order to maintain the given ratio.

Choice A is incorrect. If the width of the rectangular region increases, the length must also increase, not decrease.

Choice C is incorrect. If the width of the rectangular region increases, the length must also increase, not decrease.

Choice D is incorrect. Since the ratio of the length to the width of the rectangular region is 35 to 10, if the width of the rectangular region increases by 7 units, the length would have to increase by a proportional amount, which would have to be greater than 7 units.

Q9

D

D. 233,640

Choice D is correct. It's given that 1 furlong = 220 yards and 1 yard = 3 feet. It follows that a distance of 354 furlongs is equivalent to $354 \text{ furlongs} \times \frac{220 \text{ yards}}{1 \text{ furlong}} \times \frac{3 \text{ feet}}{1 \text{ yard}}$, or 233,640 feet.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Q10

17

The correct answer is 17. If one value is multiplied by a number, then the other value must be multiplied by the same number in order to maintain the same ratio. It's given that j is multiplied by 17. Therefore, in order to maintain the same ratio, k must also be multiplied by 17.

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Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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